

Chapter 4 Assignment

Work and Energy

K $9 \frac{3}{4}$ / 10	T $9 \frac{3}{4}$ / 10	A $3 \frac{3}{4}$ / 4	C 4 / 4
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SHOW ALL YOUR WORK IN ANSWERING QUESTIONS.

well Done!

KNOWLEDGE

1. A spring stretches by 18 mm when a 2.8 kg object is suspended from its end. How much mass should be attached to this spring so that its frequency of vibration is $f = 3.0$ Hz? Consider that the spring undergoes a simple harmonic motion.

Show all your work.

$$(1) F_s = -kx$$

$$F_s = m\ddot{y}$$

$$mg = -kx$$

$$-k = \frac{mg}{x}$$

$$-k = \frac{2.8 \cdot 9.8}{0.018}$$

$$k = 1524.44$$

$$m = 2.8 \text{ kg}$$

$$g = 9.8 \text{ m/s}^2$$

$$x = 0.018 \text{ m}$$

$$(2) T = 2\pi \sqrt{\frac{m}{k}}$$

$$T = \frac{1}{f}$$

$$\frac{1}{f} = 2\pi \sqrt{\frac{m}{k}}$$

$$\left(\frac{1}{2\pi f}\right)^2 k = m$$

$$m = \left(\frac{1}{2\pi \cdot 3}\right)^2 \cdot 1524.44$$

$$m = 4.29 \text{ kg}$$

[4 marks]

Final answer: $m = 4.29 \text{ kg}$

2. A 0.60 kg metal sphere oscillates at the end of a vertical spring. As the spring stretches from 0.12 to 0.23 m (relative to its unstrained length), the speed of the sphere decreases from 5.70 to 4.80 m/s. What is the spring constant of the spring?

Show all your work.

$$m = 0.60 \text{ kg}$$

$$x_1/h_1 = 0.12 \text{ m} \quad x_2/h_2 = 0.23 \text{ m}$$

$$v_1 = 5.7 \text{ m/s} \quad v_2 = 4.8 \text{ m/s}$$

$$\frac{1}{2}mv^2 + mgh$$

$$E_{k1} + E_{p1} = E_{k2} + E_{p2}$$

$$\frac{1}{2}mv_1^2 + \frac{1}{2}kx_1^2 = \frac{1}{2}mv_2^2 + \frac{1}{2}kx_2^2$$

$$\frac{1}{2}m(v_1^2 - v_2^2) + \frac{1}{2}k(x_1^2 - x_2^2)$$

$$(0.6 \text{ kg}) \left(9.8 \frac{\text{m}}{\text{s}^2}\right) (0.11 \text{ m})$$

$$k = \frac{-0.6(5.7^2 - 4.8^2)}{(0.12^2 - 0.23^2)}$$

$$k = 1417.27 \text{ N/m}$$

$$k = 180 \frac{\text{N}}{\text{m}}$$

Final answer: $k = 1417.27 \text{ N/m}$

$$k = 180 \frac{\text{N}}{\text{m}}$$

THINKING

1. Spiderman uses his spider webs to save a runaway train moving about 60 km/h. His web stretches a few city blocks at 500 m before the 10⁴ kg train comes to a stop. Assuming the web acts like a spring, calculate the spring constant. Show all your work.

$$v_1 = 60 \text{ km/h} = 16.7 \text{ m/s}$$

$$v_2 = 0 \text{ m/s}$$

$$m = 10^4 \text{ kg} = 10000 \text{ kg}$$

$$d/x = 500 \text{ m}$$

$$E_{\text{elastic}} = \frac{1}{2} kx^2$$

$$E_k = \frac{1}{2} mv^2$$

$$\frac{1}{2} kx^2 = \frac{1}{2} mv^2$$

$$k = \frac{mv^2}{x^2}$$

$$k = \frac{10000 (16.7)^2}{500^2}$$

$$k = 11.12 \text{ N/m}$$



[4 marks]

Final answer: $k = 11.12 \text{ N/m}$

2. A 62 kg trampoline artist jumps upward from the top of a platform with a vertical speed of 4.5 m/s.
- How fast is he going as he lands on the trampoline, 2.0 m below?
 - If the trampoline behaves like a spring of spring constant 5.8×10^4 N/m, how far down does he depress it?
- Show all your work.

a) $E_{k1} + E_{g1} = E_{k2} + E_{g2}$

$\frac{1}{2}mv_1^2 + mgh_1 = \frac{1}{2}mv_2^2 + mgh_2$

$m = 62 \text{ kg}$
 $v_1 = 4.5 \text{ m/s (up)}$
 $h_1 = 0 \text{ m}$
 $h_2 = -2.0 \text{ m}$

$\frac{1}{2}mv_1^2 + mgh_1 = \frac{1}{2}mv_2^2 + mgh_2$

$\frac{1}{2}mv_1^2 + mgh_1 - mgh_2 = \frac{1}{2}mv_2^2$

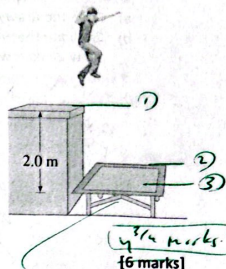
$\frac{1}{2}mv_1^2 + mgh_1 - mgh_2 = \frac{1}{2}mv_2^2$

$\frac{1}{2}mv_1^2 + mgh_1 - mgh_2 = \frac{1}{2}mv_2^2$

$v_2 = \sqrt{v_1^2 - 2gh_2}$

$v_2 = \sqrt{4.5^2 - 2(9.8)(-2)}$

$v_2 = 7.71 \text{ m/s (down)}$



b) $E_e = \frac{1}{2}kx^2$

$E_k = \frac{1}{2}mv^2$

$E_e = E_k$

$\frac{1}{2}kx^2 = \frac{1}{2}mv^2$

$x^2 = \frac{mv^2}{k}$

$x^2 = \frac{(62 \text{ kg})(7.71)^2}{5.8 \times 10^4}$

$x^2 = 0.06354$

$x = 0.252 \text{ m (down)}$

incorrect solution
 leading to incorrect
 final answer
 See Solutions!

$\frac{1}{2}mv_1^2 + mgh_1 + \frac{1}{2}kx^2 = \frac{1}{2}mv_2^2 + mgh_2 + \frac{1}{2}kx^2$

$\frac{1}{2}kx^2 + mgh_2 - \frac{1}{2}mv_2^2 = 0$

$a = \frac{1}{2}k$

$b = mg$

$c = -\frac{1}{2}mv_2^2$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$x = \frac{-mg \pm \sqrt{(mg)^2 - 4(\frac{1}{2}k)(-\frac{1}{2}mv_2^2)}}{2(\frac{1}{2}k)}$

$x = \frac{-6.267 \text{ M} \pm \sqrt{39.17 \text{ M}^2 + 1.17 \text{ M}^2}}{5.8 \times 10^4 \text{ N/m}}$

$x = 0.252 \text{ m}$

Final answer: a) $v_2 = 7.71 \text{ m/s (down)}$

Final answer: b) $x = 0.252 \text{ m (down)}$

$\Delta x = -0.263 \text{ m}$

APPLICATION

1. The drawing shows three identical springs hanging from the ceiling. Nothing is attached to the first spring, whereas a 4.50 N block hangs from the second spring. A block of unknown weight hangs from the third spring.

- a) From the drawing, determine the spring constant (in N/m).
b) Calculate the weight of the block hanging from the third spring?
Show all your work.

$$a) \Delta x = x_2 - x_1$$

$$\Delta x = 0.35 - 0.03 = 0.32 \text{ m}$$

$$F = 41.5 \text{ N}$$

$$F_s = -k\Delta x$$

$$-k = \frac{F_s}{\Delta x}$$

$$-k = \frac{-41.5 \text{ N}}{0.32 \text{ m}}$$

$$k = 30 \text{ N/m}$$

$$b) \Delta x = 50 \text{ cm} - 20 \text{ cm} = 30 \text{ cm} = 0.3 \text{ m}$$

$$k = 30 \text{ N/m}$$

$$F_s = -k\Delta x$$

$$F_s = m\vec{g}$$

$$m\vec{g} = -k\Delta x$$

$$m = \frac{-k\Delta x}{g}$$

$$m = \frac{-30 \cdot 0.3}{-9.8}$$

$$m = 0.918 \text{ kg}$$

$$\text{weight} = m\vec{g}$$

$$\text{weight} = 0.918 \cdot 9.8$$

$$\text{weight} = 8.99 \text{ N}$$

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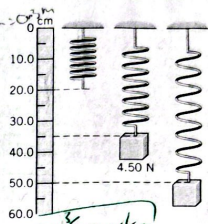
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[4 marks]

Final answer: a) $k = 30 \text{ N/m}$ Final answer: b) 8.99 N

Edwards

COMMUNICATION

1. Can the normal force on an object ever do work? Explain and provide an example.

Normal force can do work on objects if it moves ~~in a distance~~ ^{or displacement}. [2 marks]

For instance, an elevator moving up has a higher normal force pushing the people inside up, meaning it is doing work. This means it does positive work.

2. You have two springs that are identical except that spring 1 is stiffer than spring 2 ($k_1 > k_2$). If each spring is stretched using the same force, on which spring is more work done? Show your work.

Spring 2 would have more work done, because it would move more distance / displacement with the same force. [2 marks]

$$F = k\Delta x$$

$$\Delta x = \frac{F}{k}, k_1 > k_2$$

$$\therefore \Delta x_1 = \frac{F}{k_1} < \Delta x_2 = \frac{F}{k_2}$$

Spring 2 would move more

$$W = F\Delta x \cos \theta$$

$$d_2 > d_1$$

$$\therefore W_1 = F\Delta x_1 \cos \theta < W_2 = F\Delta x_2 \cos \theta$$

Spring 2 would have more work done.